Shiv S. Kammadari

Graduate Product Management | Decision-Support Systems | AI & Analytics

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Profile

SIAT graduate student with a multidisciplinary background in analytics, machine learning, software development, and engineering research. Experienced in analyzing operational data, identifying workflow bottlenecks, building decision-support dashboards, and evaluating predictive systems. Applied experience includes workflow evaluation, task-based usability studies, alpha testing, user acceptance testing (UAT), observational testing, and translating operational findings into analytics and decision-support deliverables. Across the related InRule Technology, American Express and Powerhouse workflow-testing activities, 50 internal participants took part in alpha testing and over 200 users participated across UAT, task-based usability tests, and observation combined.

Industry Experience

Senior Data Analyst | Product Analytics & Decision Support *Dallas, TX*
IT Face Inc. | Client: InRule Technology (Chicago, IL) *2024 – 2025*

- Built churn-prediction models in Python and scikit-learn with business stakeholders, contributing to a reported 15% improvement in retention-strategy effectiveness.
- Created executive Power BI dashboards for integration KPIs and operational risk, reducing manual reporting time by 50%.
- Supported a client engagement for InRule Technology, a Chicago-based enterprise software and artificial intelligence company providing decision automation, business rule management, explainable AI, and process automation tools.

Data Analyst | Market Research & Operations Analytics *Fort Worth, TX*
Powerhouse Services Inc. *2024*

- Analyzed historical work orders across 15+ U.S. states to identify bottlenecks and inform resource allocation; workflow analysis was associated with a reported 25% improvement in resource-allocation efficiency.
- Forecast service demand using SQL, Excel, and time-series analysis; improved reported forecast accuracy by 35%.
- Conducted task-based usability studies, workflow-testing sessions, and observational reviews with operational users to assess work-order information and service-planning tasks; synthesized friction points into workflow and reporting recommendations.
- Automated executive reporting workflows, reducing monthly preparation time by 60%.

Product Engineer (Automation & Analytics) *Sunrise, FL*
American Express *2023 – 2024*

- Analyzed support-case data with SQL to identify triage friction and case-categorization patterns, contributing to a reported 25% reduction in average triage time.
- Conducted task-based usability testing and UAT of support-case triage workflows, observing task completion, case-categorization decisions, and handoff friction; synthesized findings into workflow and reporting recommendations.
- Monitored machine-learning model performance and drift; built Power BI and Tableau KPI dashboards.

Industry Experience (continued)

Software Dev – Product Pipelines *Dallas, TX*
Unigo Inc. *2022 – 2023*

- Worked with product managers on tracking metrics and hypothesis evaluation for A/B-testing and predictive analytics pipelines; campaign-targeting effectiveness improved by a reported 23%.
- Maintained Azure DevOps CI/CD pipelines and developed Tableau dashboards that reduced report-generation cycles by 75%.

Data Analyst | Predictive Analytics *IN*
Eureka IT Solutions *2019 – 2020*

- Developed TensorFlow/Keras revenue-forecasting models, contributing to a reported 40% improvement in forecast accuracy; standardized data models across business units.

Research Associate | Market Intelligence *IN*
Mordor Intelligence Inc. *2017 – 2018*

- Conducted market-data analysis and time-series forecasting using Python, R, and SAS for research and client deliverables.

Applied Research

Human-in-the-Loop AI: Domain Shift & Model Explainability
University of South Florida *2021–2022*

- Studied domain shift and transferability in deep-learning medical-image analysis using NIH ChestX-ray14; adapted a ResNet-based approach into IMPACT-Net.
- Developed preprocessing, normalization, augmentation, and balanced-sampling workflows to investigate generalization across data sources.
- Integrated Grad-CAM heatmaps to inspect image regions influencing model predictions and support human review. Research code.

Master's Thesis: Experimental & Simulation Validation *BTH and DRDL | 2016 – 2017*

- Correlated physical vibration tests with finite-element simulation to validate dynamic-system behavior; thesis published in DiVA (publication).

Education

Simon Fraser University, School of Interactive Arts & Technology (SIAT) *Vancouver, BC*
M.Sc. student 2026 – Present · Graduate Entrance Fellowship recipient

University of South Florida *Tampa, FL*
M.S. Analytics and Information Systems (Analytics and ML) 2021–2022
GPA: 3.63/4.00

JNTUH–Blekinge Tekniska Högskola
Integrated Degree, Mechanical Engineering (Structural Mechanics)

Skills

Product methods: Stakeholder collaboration, workflow analysis, requirements definition, PRD structure, prioritization, acceptance criteria, metric design, prototype development, usability validation, human-in-the-loop design
Analytics & experimentation: Python, SQL, scikit-learn, TensorFlow, PyTorch, A/B testing, forecasting, model evaluation, explainable AI (Grad-CAM)
Decision-support tools: Power BI, Tableau, Excel, Git, Azure DevOps, REST APIs, MySQL, Flask, React
Systems engineering: FEA, structural dynamics, MATLAB, SolidWorks, ANSYS